

COURSE SCHEME
FOR B.TECH 8TH SEMESTER COMPUTER SCIENCE & ENGINEERING.
BATCH 2018 ONWARDS

Code	Course Name	L	T	P	Theory	Sessional	Practical	Total
HUM-812	Organizational Behavior	3	2	-	100	50	-	150
ELECTIVE-I CSE-801	A) Advanced Computer Architecture B) Data Warehousing & Data Mining	3	2	-	100	50	-	150
CSE-802	Artificial Intelligence	3	2	-	100	50	-	150
ELECTIVE-II CSE-803	A) Compiler Design B) Distributed Databases	3	2	-	100	50	-	150
CSE-804	Major Project	-	-	12	-	-	400	400
	Total	12	8	12	400	200	400	1000

CLASS: B.TECH 8th SEMESTER
BRANCH: COMPUTER SCIENCE & ENGINEERING
COURSE TITLE: ORGANIZATIONAL BEHAVIOUR
COURSE NO: HUM-812
DURATION OF EXAM: 3 HOURS

L	T	P	MARKS	
3	2	-	Theory	Sessional
			100	50

UNIT-I

Organizational Behavior: Fundamental Concepts, Nature of people, Nature of organization, Features, need to study O.B, Models of organizational behavior. Motivation-Concept & Importance, Theories of motivation – Maslow’s Need Hierarchy Theory & Herzberg’s Motivation-Hygiene Theory.

Individual behavior and its determinants: Personality-Concept & Determinants; Perception-Meaning, Definition, Perceptual Process, internal & external factors in perceptual selectivity; Attitude-Features, Components, Formation of Attitudes; OB Modification: Steps in OB Mod & Utility of OB Mod

UNIT-II

Communication: Characteristics, Process, Types, Barriers & Breakdowns in Communication, Overcoming Communication Barriers, Guidelines for Effective Communication.

Organization as a Social system: Concept, Characteristics & Objectives of Organization Development, Organization Development Process. Organizational Change: Nature of Organizational Change, Forces to Change, Causes of Resistance to Change, Techniques of overcoming Resistance to Change, Response & Reactions to Change

UNIT-III

Leadership and Organization Development: Leadership–Characteristics and Functions of Leader, Qualities of a good Leader, Importance of leadership, Styles of Leadership; Dynamics of Conflict – Nature of Conflict, Types of Conflict, Stages in conflict, Resolution of conflicts.

UNIT-IV

Culture & Management with Indian reference: Meaning, definition & Elements of culture, Characteristics of organizational culture, Functions of Culture in Management, Basic Foundations of Indian Culture and its linkage with the practice of management in India.

BOOKS RECOMMENDED:

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|---|--------------------------|---------------------------------|
| 1 | Organizational Behavior | - John W. Newstrom& Keith Davis |
| 2 | Organizational Behavior | - Robbins, - L. M. Prasad |
| 3 | Management | - G. S. Sudha |
| 4 | Principles of Management | - Dr. NeeruVaisisth |

COURSE OUTCOMES COURSE NAME:ORGANIZATIONAL BEHAVIOUR COURSE NO. HUM-812 After learning this course Students will be able to:	
CSE812A.1	Remember and understand work in organizations by gaining proper knowledge about concepts, models, and theories of O.B and detail knowledge about personality, perception etc.
CSE812A.2	Apply and transmit their ideas, facts, thoughts, feelings, and values effectively through effective communication tools.
CSE812A.3	Analyze and cope up with the organizational changes in future.
CSE812A.4	Evaluate the work for development of the organizations.
CSE812A.5	create to work as an effective leader for organizational development and organizational change

CLASS: B.TECH 8th SEMESTER
BRANCH: COMPUTER SCIENCE & ENGINEERING
COURSE TITLE: ADVANCED COMPUTER ARCHITECTURE
COURSE NO: CSE-801(A) (ELECTIVE-I)
DURATION OF EXAM: 3 HOURS

L	T	P	MARKS	
3	2	-	Theory	Sessional
			100	50

UNIT-I

Introduction to Parallel Processing: Parallel Processing, Architectural classification schemes, Parallelism in Uniprocessor systems, Types of Parallelism.

Pipelining Processing: Introduction to pipelining, Classification of Pipeline Processors, General pipelines and reservation tables, Principles of Designing pipeline processors, Pipeline hazards and conflicts.

UNIT-II

Vector and SIMD Array Processing: Introduction to Vector Processing, Classification of vector Processing, Vector computing process, Characteristics of vector Processing, Array Processors, Classification of Array Processors, Masking and Data Routing Mechanisms.

UNIT-III

Multiprocessor Systems: Introduction to multi processors, Classification of multi processors systems, Processor characteristics for multiprocessing, Multi-processor execution modes, Multi-processor operating system, Interconnected Networks for multiprocessors and Multi-computer, Performance evaluation methods.

UNIT-IV

Parallel Memory Organization: Memory organization in multi-processor systems, Interleaved Memory Organization, Characterization and organization of cache memory, Multi-cache Problems, Virtual memory concepts and techniques, input- output organization, Characteristics of input-output systems, input- output processor. **Parallel computer models and Program Parallelism:** Models of parallel computers, Introduction to Parallel Algorithms, Parallel Programming Paradigms, Issues in implementing algorithms on Parallel Computers.

BOOKS RECOMMENDED:

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|---|---|
| 1. Computer Architecture: A Qualitative Approach | Hennessey & D.A. Patterson. |
| 2. Computer Architecture: Pipelined & Parallel Processor Design | Michael J. Flynn. |
| 3. Advanced Computer Architecture | Kai Hwang & Briggs. |
| 4. Foundations of Parallel Processing | R.K.Ghose, RajanMoona and Phalguni Gupta. |

COURSE OUTCOMES

COURSE NAME:ADVANCED COMPUTER ARCHITECTURE

COURSE NO. CSE-801(A)

After learning this course Students will be able to:

CSE801(A).1	Remember the Latest development in computer architecture.
CSE801(A).2	Understand parallel processing/computing basic terminologies.
CSE801(A).3	Apply and Acquire knowledge about the latest concepts & techniques in pipelining processing & hazards.
CSE801(A).4	Analyze performance evaluation methods in multiprocessor & multicomputer.
CSE801(A).5	Create virtual memory & memory interleaving in PC.

CLASS: B.TECH 8th SEMESTER
BRANCH: COMPUTER SCIENCE & ENGINEERING
COURSE TITLE: DATA WAREHOUSING & DATA MINING
COURSE NO: CSE-801(B) (ELECTIVE-I)
DURATION OF EXAM: 3 HOURS

L	T	P	MARKS	
3	2	-	Theory	Sessional
			100	50

UNIT-I

Introduction to Data Warehousing & Data Mining: Need for data Warehousing & Data Mining, Data Mining on different kinds of Data, Data Mining Functionalities - Characterization and Discrimination, Association Analysis, Classification and Prediction, Cluster Analysis, Outlier Analysis, Evolution Analysis, Classification of Data Mining System, Data Mining Applications.

UNIT-II

Data Warehouse and OLAP Technology for Data Mining: Differences between Operational database systems and Data warehouse, Multidimensional Data Model- Data Cubes, Star, Snowflakes & Fact Constellations Schemes, Concept Hierarchies, OLAP Operations, Query Model, Data Warehouse, 3-Tier Architecture, Types of OLAP Servers: ROLAP, MOLAP, HOLAP, Data Warehouse Usage, On-Line Analytical Processing to On-Line Analytical Mining, Data Marts, Meta Data.

UNIT-III

Data Preprocessing and Mining: Data Cleaning, Data Integration & Transformation, Data Reduction, Discretization and Concept Hierarchy Generation. Kinds of Knowledge to be Mined, Concept Hierarchies, Interesting measures, Presentation & Visualization of Discovered Patterns.

UNIT-IV

Concept Description and Association Rules Mining: Concept Description, Data Generalization and Summarization based Characterization, Analytical Characterization, Attribute Relevance Analysis. Association Rules Mining Basic Concepts, Mining Single Dimensional Boolean Association Rules from Transactional databases- Apriori Algorithm. Introduction to classification and Prediction: Basic Concepts, Classification by Decision Tree Induction, Linear and multiple Regression, Non Linear Regression.

BOOKS RECOMMENDED:

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|--|------------------------------|
| 1. Data Mining: Concepts & Techniques | Jiawei Hun, Micheline Kamber |
| 2. Modern Data Warehousing Mining & Visualization Core Concept | George M. Marakas |
| 3. Data Mining | Pieter Adrians |
| 4. Data Mining with Microsoft SQL Server | Seidman |

COURSE OUTCOMES

COURSE NAME: DATA WAREHOUSING & DATA MINING

COURSE NO. CSE-801(B)

After learning this course Students will be able to:

CSE801(B).1	Remember the importance of pre-processing the given datasets
CSE801(B).2	Understand the concepts of data warehouse and data mining
CSE801(B).3	Apply data pre-processing techniques to build data warehouse
CSE801(B).4	Analyze transaction databases for association rules.
CSE801(B).5	Evaluate methods for outlier analysis and develop skills to design data warehouse for an enterprise.

CLASS: B.TECH 8th SEMESTER
BRANCH: COMPUTER SCIENCE & ENGINEERING
COURSE TITLE: ARTIFICIAL INTELLIGENCE
COURSE NO: CSE-802
DURATION OF EXAM: 3 HOURS

L	T	P	MARKS	
3	2	-	Theory	Sessional
			100	50

UNIT-I

Artificial Intelligence: - The AI problems, AI techniques, The level of the model, criteria for success, AI tasks. Problems, problem spaces & Research: - Defining the problem as a space, search, production system problem characteristics, Production system characteristics, Issues in the design of search programs, two path problems.

UNIT-II

Symbolic reasoning under uncertainty: -Introduction to non - monotonic Reasoning, Logics for non-monotonic reasoning, Implementation Issues, Augmenting a Problem Solver. Implementation by: a) Depth - First Search b) Breadth - First Search. Statistical reasoning: - Probability & Bayes Theorem, Certainty Factors & Rules Based Systems, Bayesian networks, Dempster Shafer Theory, Fuzzy logic, Introduction to Expert System development.

UNIT-III

Using Predicate Logic: -Representing simple facts, its logic representing instances and its Functions and Relationships, Computable Predicates, Resolution, Natural Deduction, Conversion to Clause Form.

Knowledge Representation Issues: - Representation and mappings, approaches to knowledge representation, Issues of knowledge representation, the frame problem, Semantic networks.

UNIT-IV

Representing knowledge using rules: -Procedural Vs Declarative Knowledge, Logic Programming Forward Vs Backward Searching, Matching, Control Knowledge. Heuristic Search Techniques: - Generate & test, Hill Climbing, Best First Search, Problem reduction, Constraint, Satisfaction, Means and analysis.

BOOKS RECOMMENDED:

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|----|---|--------------------------|
| 1. | Artificial Intelligence | Elaine Rich Kevin Knight |
| 2. | Principles of A.I Expert system development | David W. Rolston. |

COURSE OUTCOMES

COURSE NAME:ARTIFICIAL INTELLIGENCE

COURSE NO. CSE-802

After learning this course Students will be able to:

CSE802.1	Remember and understand the concept and fundamental methods of AI.
CSE802.2	Apply basic AI algorithms to solve the problems.
CSE802.3	Analyses how uncertainty is being tackled in knowledge representation and reasoning process.
CSE802.4	Evaluate the concept of structural representation of knowledge using rules.
CSE802.5	Create fuzzy logic to implement expert systems.

CLASS: B.TECH 8th SEMESTER
BRANCH: COMPUTER SCIENCE & ENGINEERING
COURSE TITLE: COMPILER DESIGN
COURSE NO: CSE-803(A) (ELECTIVE-II)
DURATION OF EXAM: 3 HOURS

L	T	P	MARKS	
3	2	-	Theory	Sessional
			100	50

UNIT-I

Introduction–Languages Processors, the typical structure of a Compiler.

Programming Language –High level programming languages, definition of programming languages, the syntax and semantics of basic data and control structures in high level programming languages.

Lexical analysis –Role of Lexical Analyzer, input buffering, a simple approach to Design of Lexical Analyzers, Regular Expressions, Finite Automata, Regular expression to Finite Automata, Conversion of NFA to DFA, Minimizing the number of states of a DFA.

UNIT-II

The Syntactic Specification of Programming Languages –Definition of Grammars (Context free grammar), derivation, parse tree, ambiguity, non-context free language constructs.

Basics Parsing Techniques –Parsers- Shift reduce parsing, Operator precedence parsing, top -down parsing, Predicative parsers, LR parsers.

Syntax directed translation- Syntax directed translation schemes. Implementation of syntax directed translators.

UNIT-III

Intermediate code Generation - Intermediate code, postfix notation, three address code-quadruples triples, translation of Assignment statement, Boolean Expression, Statements that alter the flow of control.

Symbol Table Organization –The content of symbol table, Data structure of symbol table

Run-Time memory Allocation-Static and Dynamic memory allocation, Static allocation of space

– Activation trees, activation records, Procedure calls, parameter passing.

UNIT-IV

Error Detection and Recovery-Errors, lexical phase errors, syntactic phase errors, semantic errors.

Code optimization- Loop optimization, DAG Representation of basic blocks, Global data flow Analysis.

Code generation- Issues in the design of code generator, Peephole optimization, a simple code generator Register Allocation & Assignment.

BOOKS RECOMMENDED:

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|--|---|
| 1. Principles of compiler design | Alfred V.Aho, Jeffrey D Ullman |
| 2. Principles of compiler design | Aho v. Ullman, Sethi |
| 3. Theory of parsing Translation & Compiling | Aho. ullman |
| 4. Compiler construction | MunishJha |
| 5. Compilers Principles, Techniques & Tools | Alfred V. Aho, Monika S Lam, Ravi Sethi, Jeffrey D Ullman |

COURSE OUTCOMES

COURSE NAME:COMPILER DESIGN

COURSE NO. CSE- 803(A)

After learning this course Students will be able to:

CSE803(A).1	Remember the functionality of each phase involved in Compilation process and understand intermediate code representations .
CSE803(A).2	Apply the parsing techniques including Bottom-up and Top-down parsing for the given programming construct described in Context Free Grammar.
CSE803(A).3	Analyze the concepts of storage administration for different programming environments.
CSE803(A).4	Evaluate different error recovery routines to recover the errors seen at different phases of compilation.
CSE803(A).5	Create the machine code by considering all the functionalities involved in different phases of the compilation process.

CLASS: B.TECH 8th SEMESTER
BRANCH: COMPUTER SCIENCE & ENGINEERING
COURSE TITLE: DISTRIBUTED DATABASES
COURSE NO: CSE-803(B) (ELECTIVE-II)
DURATION OF EXAM: 3 HOURS

L	T	P	MARKS	
3	2	-	Theory	Sessional
			100	50

UNIT-I

Transaction and schedules, concurrent execution of transaction, conflict and views serializability, testing for serializability, concepts in recoverable and cascadeless schedules.
Lock based protocols, time stamp based protocols, multiple granularity and multiversion techniques, enforcing serializability for locks, locking system with multiple lock modes, architecture for locking scheduler.

UNIT-II

Distributed transaction management, data distribution, fragmentation and replication techniques, distributed commit, distributed locking schemes, long duration transactions, moss concurrency protocol.

UNIT-III

Issues of recovery and atomicity in distributed database, traditional recovery technique, log based recovery, recovery with concurrent transactions, recovery in message passing systems, checkpoints, algorithm for recovery line, concepts in orphan and inconsistent messages.

UNIT-IV

Distributed query processing, multiway joins, semi joins, cost based query, optimization for distributed database, updating replicated data, protocols for distributed deadlocks detection, eager and lazy replication techniques.

BOOKS RECOMMENDED:

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|---|---------------------------------|
| 1. Principles of distributed database systems | M Tamer Ozsu Patrick Valduriez. |
| 2. Distributed database system | David A. Bell, Jane B. Grimson |
| 3. Managing Distributed Database | Donald K. Burleson |
| 4. Advanced Distributed Systems | Felix F. Romas |

COURSE OUTCOMES

COURSE NAME: DISTRIBUTED DATABASE

COURSE NO. CSE-803(B)

After learning this course Students will be able to:

CSE803(B).1	Remember conceptual understanding of Distributed Systems
CSE803(B).2	Understand different techniques for data fragmentation, replication, and allocation during the distributed database design process.
CSE803(B).3	Analyze simple strategies for executing a distributed query to select the strategy that minimizes the amount of data transfer.
CSE803(B).4	Evaluate how two-phase commit protocol is used to deal with committing a transaction that accesses databases stored on multiple nodes
CSE803(B).5	Design distributed concurrency control system based on the distinguished copy techniques and the voting methods.

CLASS: B.TECH 8th SEMESTER
BRANCH: COMPUTER SCIENCE & ENGINEERING
COURSE TITLE: MAJOR PROJECT
COURSE NO: CSE-804
DURATION OF EXAM: 3 HOURS

L	T	P	MARKS
-	-	12	400

The student will complete their assigned project work initiated in 7th semester under course No.CSE-708 and submit a detailed project report individually to the Head of the department.

Guidelines for evaluation of Project work in 8th semester:

There shall be a mid-semester evaluation, followed by an End Semester (Final) Evaluation

Sub-distribution of marks:

- For External Examiner : 100
- For Internal Examiner : 300

Sub-distribution of internal Marks:

- Mark distribution of internal Project work as per the University statutes shall be based on:

a.	Viva-Voce	=	90	30%
b.	Presentation	=	90	30%
c.	Report	=	120	40%
Total		=	300	

COURSE OUTCOMES

COURSE NAME:MAJOR PROJECT

COURSE NO. CSE-804

After learning this course Students will be able to:

CSE804.1	Complete their assigned project work initiated in minor project.
CSE804.2.	Demonstrate the project work followed by question-answer session
CSE804.3.	Present and submit the detailed project report.